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Charged Block Copolymers: From Fundamentals to Electromechanical Applications

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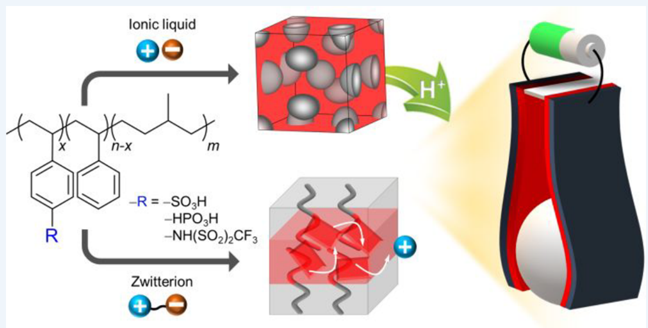

CONSPECTUS: Charged block copolymers are promising materials for next-generation battery technologies and soft electronics. Although once it was only possible to prepare randomly organized structures, nowadays, well-ordered charged block copolymers can be prepared. In addition, theoretical and experimental analyses of the thermodynamic properties of charged polymers have provided insights into how to control nanostructures via electrostatic interactions and improve the ionic conductivity without compromising mechanical strength, which is crucial for practical applications. In this Account, we discuss methods to control the self-assembly and ion diffusion behavior of charged block copolymers by varying the type of tethered ionic moieties, local concentration of embedded ions with controlled electrostatic interactions, and nanoscale morphology. We discuss with particular emphasis on the structure−transport relationship of charged block copolymers using various ionic additives to control the phase behavior electrostatically as well as the ion transport properties. Through this, we establish the role of interconnected ionic channels in promoting ion-conduction and the importance of developing three-dimensional interconnected morphologies such as gyroid, orthorhombic *Fddd* (O^{70}) networks, body-centered cubic (bcc), face-centered cubic (fcc), and A15 structures with well-defined interfaces in creating less tortuous ion-conduction pathways. Our prolonged surge and synthetic advances are pushing the frontiers of charged block copolymers to have virtually homogeneous ionic domains with suppressed ion agglomeration via the nanoconfinement of closely bound ionic moieties, resulting in efficient ion conduction and high mechanical strength.

Subsequently, we discuss how, by using zwitterions, we have radically improved the ionic conductivity of single-ion conducting polymers, which have potential for use in next-generation electrochemical devices owing to the constrained anion depletion. Key to the improvement stems from hierarchically ordered ionic crystals in nanodomains of the single-ion block copolymers through the self-organization of the dipolar/ionic moieties under confinement. By precisely tuning the distances between ionic sites and the dipolar orientation in the ionic domains with varied zwitterion contents, unprecedented dielectric constants close to those of aqueous electrolytes have been achieved, leading to the development of high-conductivity solid-state single-ion conducting polymers with leak-free characteristics. Further, using these materials, low-voltage-driven artificial muscles have been prepared that show a large bending strain and millisecond-scale mechanical deformations at 1 V in air without fatigue, exceeding the performance of previously reported polymer actuators. Finally, smart multiresponsive actuators based on tailor-made charged polymers capable of programmable deformation with high force and self-locking without power consumption are suggested as candidates for use in soft robotics.

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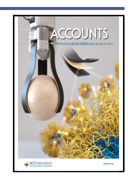

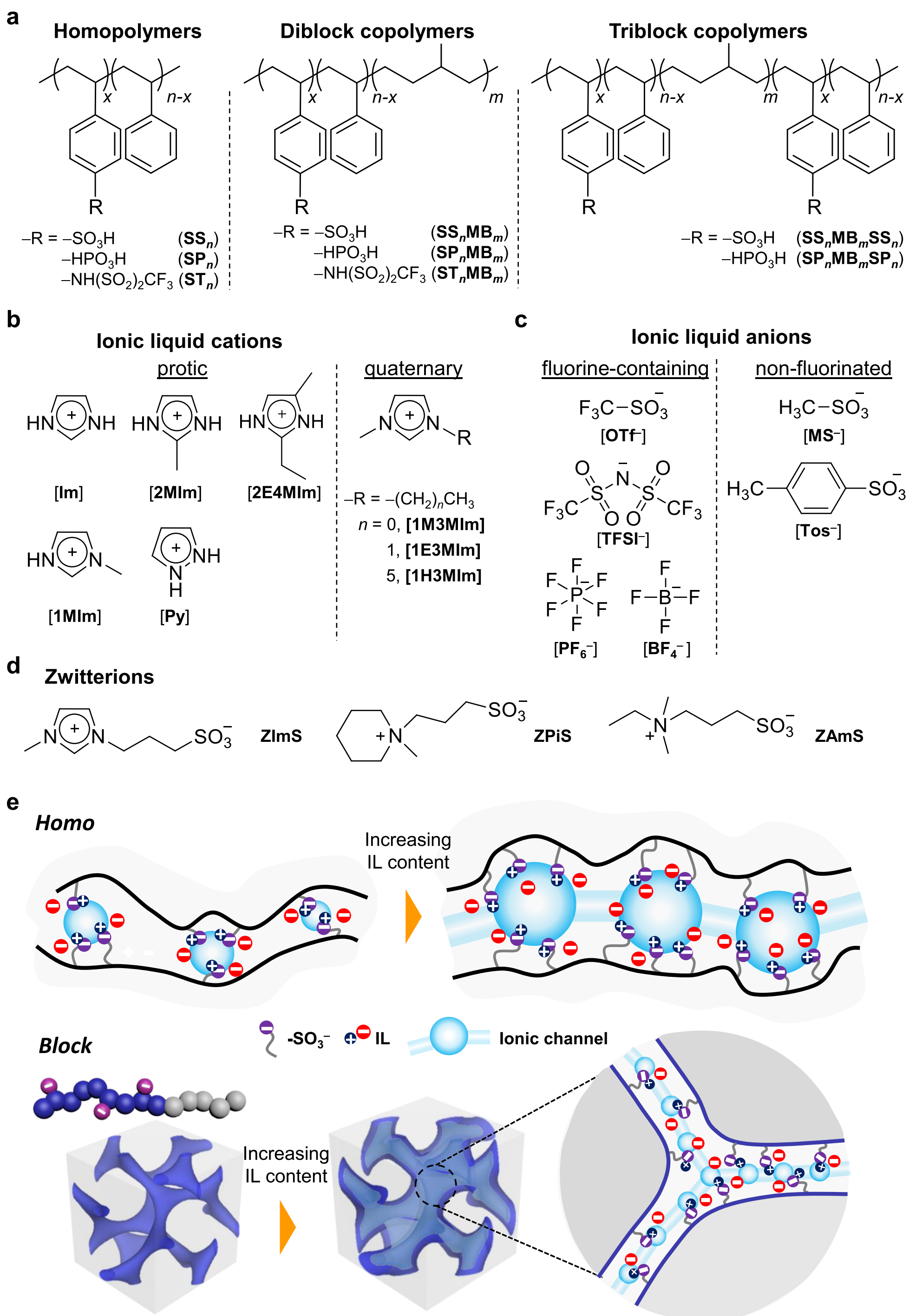

Figure 1. Chemical structures of (a) homopolymers and block copolymers tethered with various acid moieties, (b) IL cations, (c) IL anions, and (d) zwitterions. (e) The formation of ionic channels via percolation of ion clusters by embedding ILs into acid-tethered homo- and block copolymers.

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INTRODUCTION

Increasing demand and advances in energy storage and conversion technologies have drawn attention toward polymer electrolytes because of their easy processability, versatility, adaptability, nonvolatility, and low cost.[4−7] Polymer electrolytes are used in fuel cells[8−10] and supercapacitors,[11] and their use in other energy technologies and soft electronics is growing. An important property of polymer electrolytes is ion transport, which can be enhanced using phase-separated morphologies containing percolated ion clusters.[12,13] Thus, their phase behavior, ion transport, and morphological factors, as well as the relationship between these factors, must be studied.[14−16] Recently, synthesis and improved conductivity of rationally designed charged polymers have been reported, but

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this enhanced conductivity comes at the cost of mechanical strength.[17]

Charged block copolymers overcome this trade-off between ion transport and mechanical robustness by exploiting nanoscale self-assembly, and these intriguing materials have applications in next-generation electrochemical technologies.[1,3,18] However, the thermodynamics related to charged block copolymers is complex as a result of Coulombic interactions between charged moieties in the polymer matrix. Several key experimental and theoretical investigations of the self-assembly of charged block copolymers have been published recently,[19−21] but the synthesis of well-defined morphologies remains challenging. In this Account, we provide an overview of state-of-the-art block copolymer electrolytes and suggest future research directions.

We first discuss our methods to improve the ion-transport efficiency of block copolymer electrolytes, even those with nonionic blocks, by developing well-defined ionic domains.[1] We also discuss the structure−property relationship by studying acid-tethered block copolymers in which the types and local concentrations of acid moieties are quantitatively controlled by postmodification approaches.[14,22,23] Further, we show how the morphologies and ion transport properties of acid-tethered block copolymers can be systematically varied using ionic additives, ranging from ionic liquids (ILs) to zwitterions, yielding unexpected thermodynamic effects and enabling control over the curvature and interfacial features of the ionic domains.[3,22−29] Consequently, polymer electrolytes with excellent ion conductivity, mechanical strength, heat endurance, and electrochemical stability are reported.

We also present the unprecedented phase behavior related to the strength of intermolecular interactions in ionic domains and enhancement of ionic conductivity by creating hierarchically ordered ionic crystals through nanoconfinement.[3] We finally discuss low-voltage artificial muscles and soft robotics using charged block copolymers: so-called ionic electroactive polymer (*i*EAP) actuators.[2,30−33] Through our research, we hope to bridge the gap between the fundamental science and application of these functional polymers in electrochemical devices.

Acid-Tethered Polymers Containing Ionic Additives

Extensive research into random ionomers having ammonium,[34] sulfonate,[35] phosphonate,[36] and imidazolium moieties[37] has been conducted in the last 50 years. Eisenberg et al.[38,39] reported that ion clustering in charged polymers is induced by strong electrostatic interactions between ions surrounded by low-dielectric polymer backbones. Percolation of ionic clusters with effective long-range interactions is crucial for achieving high-conductivity polymer electrolytes but, paradoxically, these interactions reduce the segmental dynamics and ion-transport efficiency.[40,41] The study of the self-assembly and ion/dipole interplay at the molecular level is also challenging because of the random distribution of ionic/dipolar entities in most ionomers.

To address these challenges and design model charged polymers having well-connected ionic clusters, we have explored acid-tethered polymers. Specifically, to control the ion clustering, we experimented with changing the acid moiety, local concentration of ions, and phase-separated structures. Figure 1a shows the chemical structures of acid-tethered polymers in homo and block architectures using poly(styrenesulfonate) (SS_n), poly(styrenephosphonate) (SP_n), and poly(styrenesulfonyl(trifluoromethanesulfonyl)imide) (ST_n), and poly(methylbutylene) (MB_m) as hydrophobic blocks.[1−3,14,22−31,33,42]

Further, instead of using water or liquid electrolytes, we introduced various ILs to the above-mentioned polymers, thus increasing thermal/electrochemical stability.[43,44] Owing to the noncovalent cation−anion interactions in ILs, we could prepare various cation and anion combinations that can interact with the polymer. Depending on the position of the alkyl substituent at heterocycles, Figure 1b,[24,27,29,30] these ionic components are classified as protic or quaternary having different intermolecular interactions with the polymer matrix. Overall, protic and less-substituted cations show strong binding affinities with the tethered acid moieties because acid prefers to have contact with N−H sites of heterocycles, and the resultant electrostatic interactions have strong directionality.[27,45]

Figure 1c shows the chemical structures of our IL anions, which affect the solvation energies in the polymer matrix.[1,24,26,30] In addition to this, the anion has a profound effect on the morphologies and mechanical properties of IL-containing charged block copolymers, and the roles of the IL cations and anions in modulating the physicochemical properties of these polymers will be discussed later. We have also used three zwitterions as dipolar, neutral additives (Figure 1d) to control the dielectric environment around the tethered ions, facilitating ion dissociation.[2,3]

Ion transport in IL-containing acid-tethered polymers occurs along ionic channels, whose connectivity and local ion concentrations are tunable. Figure 1e depicts the formation of ionic clusters after embedding ILs in acid-tethered homo- and block copolymers, as well as the formation of ionic channels via percolation at increased IL contents. Increasing the concentration of tethered acid moieties can enhance proton hopping because protic sites are close, but this increases the polymer glass transition temperature (T_g).[46] Increasing the IL content does not always increase the ion-transport efficiency because of ion dilution and ionic channel disruption by swelling. Furthermore, given that a high IL content reduces the mechanical strength, increasing the ionic conductivity and mechanical properties simultaneously remains challenging.

Unlike homopolymers, block copolymers have an inherent advantage: nanoscale self-assembly is possible. Thus, the ionic channels can be tailored by varying the thermodynamic factors, specifically, the degree of polymerization (N), Flory−Huggins interaction parameter (χ), and block volume fraction (f).[47] By confining ionic moieties in nanoscale ionic domains, ion percolation is enhanced, even at low IL contents (Figure 1e).[1,48] As reported earlier, ionic conductivity of block/graft copolymers could increase, compared to random copolymer analogues.[49,50] However, the morphologies of most charged copolymers are ill-defined, suffer from ion aggregation, and are process-dependent at high ion concentrations.

Morphology−Transport Relationship of Acid-Tethered Block Copolymers

In 2010, we developed high-conductivity nanostructured polymer electrolytes by embedding imidazolium ILs into poly(styrenesulfonate)-*b*-poly(methylbutylene) (SS_nMB_m) block copolymers by varying sulfonation levels (SLs), N, and IL content.[1] Compared to SS_n homopolymer having the same IL content, SS_nMB_m has a conductivity a few orders of magnitude higher under anhydrous conditions, even better

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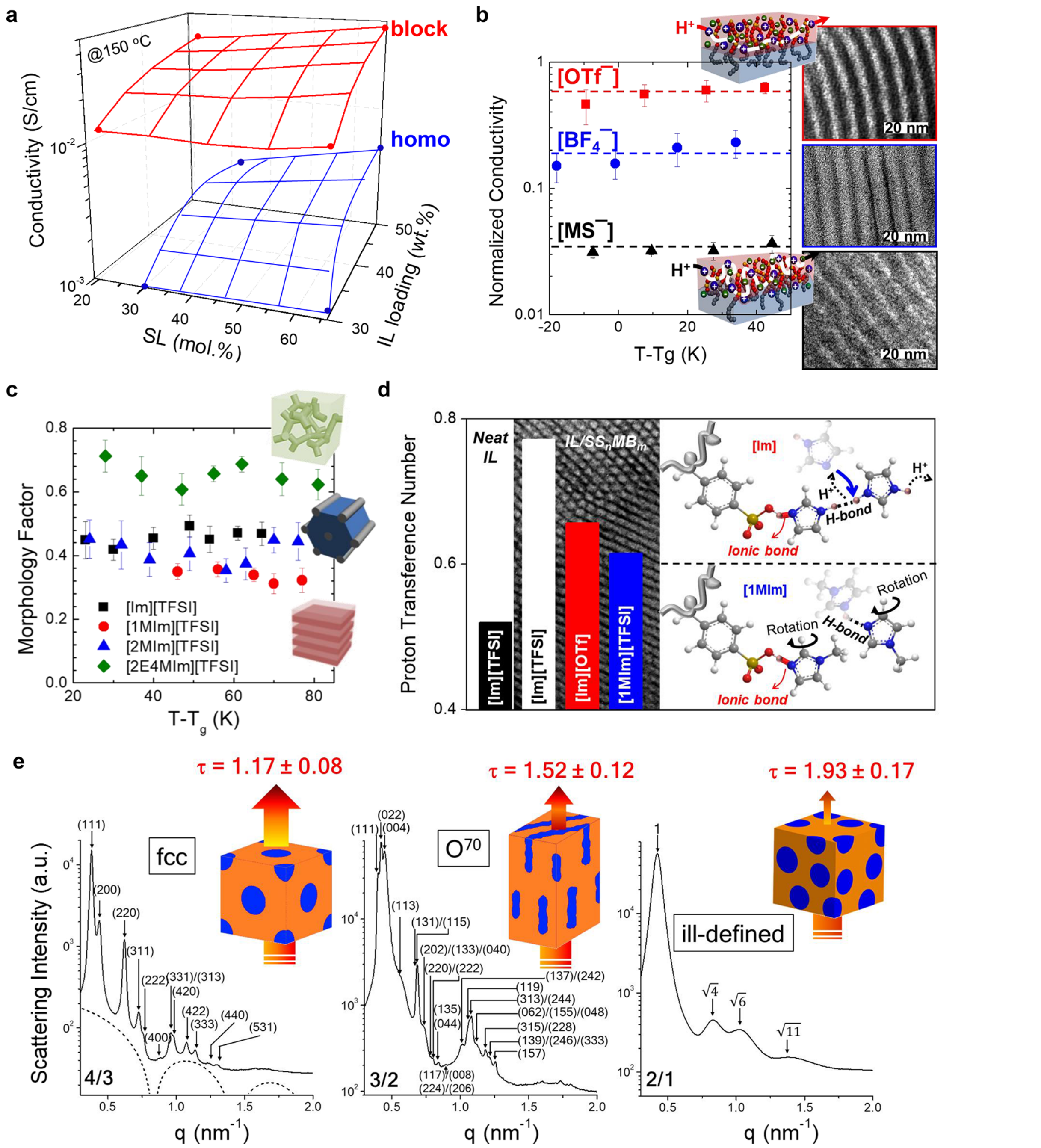

Figure 2. (a) Improved conductivity of SS_nMB_m containing ILs, compared to SS_n counterparts. Reproduced with permission from ref 1. Copyright 2010 Springer Nature. (b) Role of IL anions in determining ion transport efficiency, (c) morphology-transport relationship, and (d) proton transference number of SS_nMB_m containing ILs. Reproduced with permission from refs 26, 27, and 29, respectively. Copyrights 2013, 2012, and 2014 American Chemical Society. (e) Building more efficient ion-conduction paths for SS_nMB_m with nonstoichiometric ILs. Reproduced with permission from ref 28. Copyright 2016 American Chemical Society.

than that of Nafion, which had the highest reported conductivity at that that time. Figure 2a shows representative data obtained from $SS_{50}MB_{73}$ and SS_{60} containing [1M3MIm]-[MS], displaying facilitated proton transport in well-defined hexagonal (HEX) cylinder or perforated lamellar nanodomains of $SS_{50}MB_{73}$ with reduced activation energy.

We next fine-tuned the solvation energies of the ILs in SS_nMB_m with various anions (OTf^-, MS^-, and BF_4^-) but the same cation ($1E3MIm^+$).[26] Similar lamellar (LAM) structures were formed for $SS_{30}MB_{44}(17)$ having very different ion-transport efficiencies (by several orders of magnitude) (Figure 2b).[26] Transmission electron microscopy (TEM) and X-ray reflectivity data indicated that the chemical affinities of IL anions and SS chains affected the SS_n/MB_m interfacial roughness. For example, the swelling of SS_n domains and sharp interfacial width of the LAM structures containing OTf^- anions created efficient ion-conduction pathways, whereas the structures containing MS^- anions contained tortuous ones because of interfacial mixing. In contrast, [1E3MIm][BF_4] contained thinner SS_n domains because of the poor thermodynamic miscibility but a higher normalized conductivity than the [1E3MIm][MS] analogues, indicating that

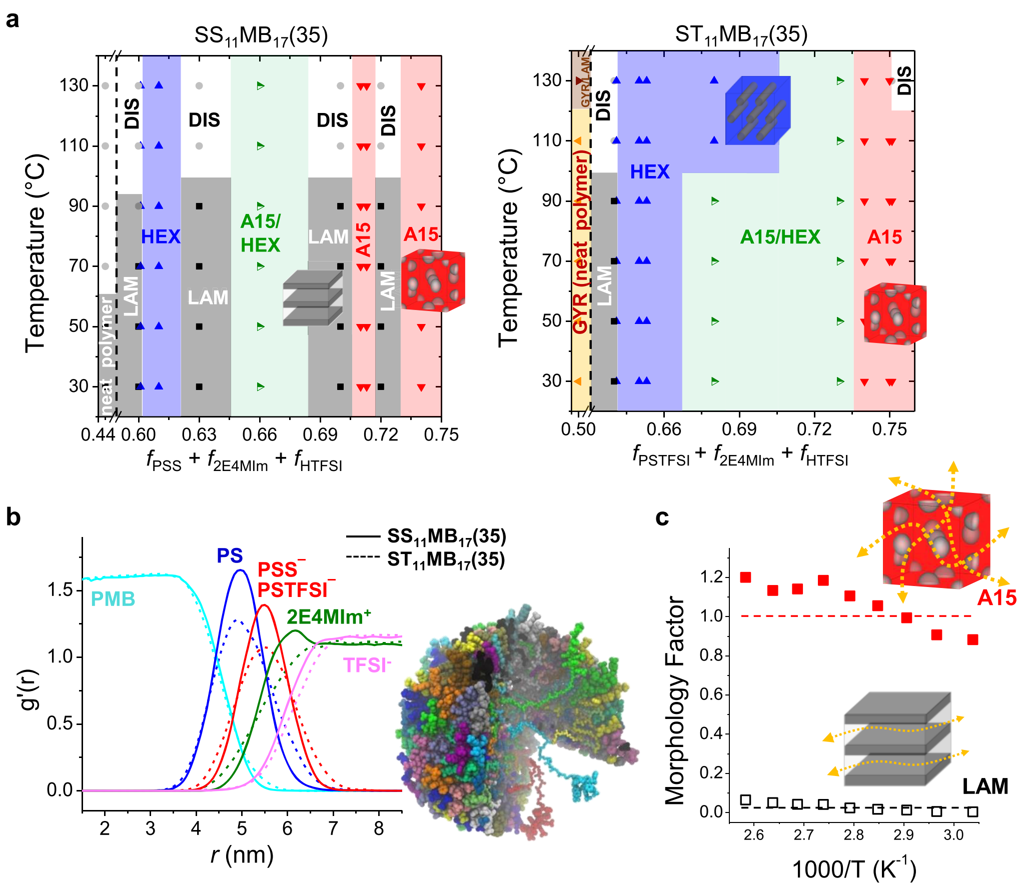

Figure 3. (a) Phase diagrams of $SS_{11}MB_{17}(35)$ and $ST_{11}MB_{17}(35)$ containing nonstoichiometric 2E4MIm/HTFSI. (b) MD simulation snapshot and radial distribution of charged block copolymer micelle dissolved in 2E4MIm/HTFSI. (c) Morphology factors of $SS_{11}MB_{17}(35)$ samples having A15 and LAM structures, calculated using the SAX−Ottino equation. Reproduced with permission from ref 23. Copyright 2021 The Authors. Published by PNAS. Distributed under a Creative Commons Attribution License (CC BY-NC-ND 4.0) https://creativecommons.org/licenses/by-nc-nd/4.0/.

interfacial properties dominate the ion-transport in charged block copolymers.

As mentioned earlier, because thermodynamics, morphology, and polymer−IL interactions are interrelated, assessing how they affect ionic conductivity is challenging. To quantify the structure−transport relationship, we also used nonstoichiometric imidazolium ILs with $TFSI^-$ anion.[27−29] The slight modification of imidazolium ring structures resulted in changes to the morphologies and segregation strengths of the IL-containing SS_nMB_m because of the different electrostatic interactions of the cations with the charged SS chains. For example, for $SS_{30}MB_{40}(17)$ with 30 wt % [Im][TFSI] and [1MIm][TFSI], highly segregated HEX structures were obtained because $-SO_3^-$ binds strongly with Im^+ and $1MIm^+$ cations.[27] In contrast, weakly segregated LAM structures formed with [2MIm][TFSI] having weakly bound $2MIm^+$ cations. Further modulation of the interactions with disubstituted $2E4MIm^+$ cations resulted in a well-defined gyroid structure. Using small-angle X-ray scattering (SAXS) and Ottino's effective medium theory to analyze the conductivity results, we calculated morphology factors ($\sigma/\phi\sigma_{max}$; σ is block copolymer conductivity, σ_{max} is homopolymer conductivity, and ϕ is volume fraction of ionic domains in block samples) of 0.6−0.7 for the gyroid samples, unlike the low value of 0.4 for the LAM and HEX counterparts (Figure 2c). Given that the morphology factor has been regarded as a quantitative indicator of how morphological defects (grain boundaries, dead ends, interfacial mixing) of block copolymer electrolytes contribute to conductivity degradation, this indicated more efficient ion transport across the network morphology.

As shown in Figure 2d, the proton transference numbers (measured by pulsed-field gradient spin−echo NMR) revealed the dependency on the type of IL cation and anion, for which the highest value was 0.77 for $SS_{37}MB_{54}(32)$ containing [Im][TFSI]. This reflects the facilitated diffusion of protons across the SS_n nanodomains via hydrogen-bonding interactions with Im^+ cations rather than an effect of the IL alone, which yielded a low value of 0.52.[29] In fact, on changing Im^+ to $1MIm^+$ (that is, fewer protic sites), the proton transference number fell to 0.61 because the rotation of [1MIm] is required to achieve proton hopping. Although IL-loaded polymer membranes had been reported before, the use of protic IL-loaded polymers and the modulation of long-range interactions in nanoscale ionic domains have been rarely reported.

We systematically varied the nonstoichiometry of protic ILs, Figure 2e, to yield new three-dimensional nanostructures such as fcc (4/3), O^{70} (3/2), and disordered cubic (2/1) phases when using 2E4MIm/HTFSI. The largely deviated tortuosities (τ) suggest the significance of preparing tailor-made

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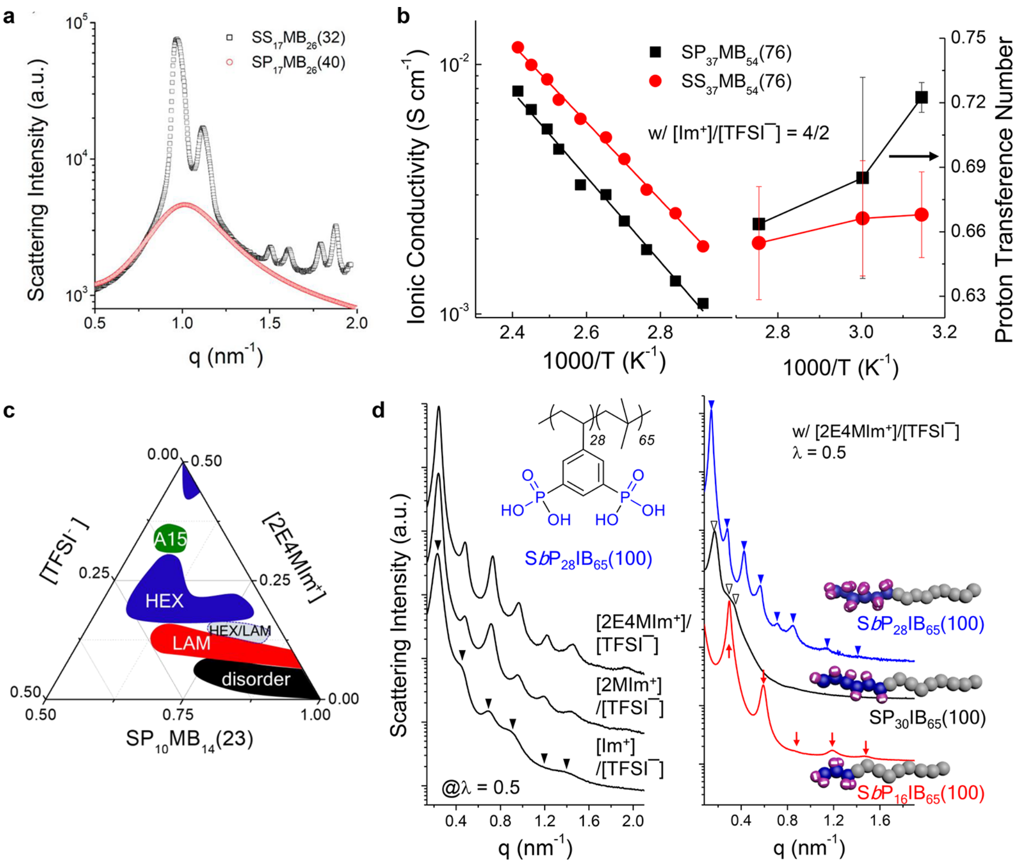

Figure 4. (a) SAXS profiles of $SS_{17}MB_{26}(32)$ and $SP_{17}MB_{26}(40)$. (b) Ionic conductivity and proton transference number for $SS_{37}MB_{54}(76)$ and $SP_{37}MB_{54}(76)$ containing nonstoichiometric Im/TFSI. Reproduced with permission from ref 14. Copyright 2015 American Chemical Society. (c) Phase diagram of $SP_{10}MB_{14}(23)$ containing nonstoichiometric 2E4MIm/TFSI. Reproduced with permission from ref 25. Copyright 2016 Royal Society of Chemistry. (d) SAXS profiles of $SbP_{28}IB_{65}(100)$ containing different ILs (left panel) and those of $SP_{30}IB_{65}(100)$, $SbP_{28}IB_{65}(100)$, and $SbP_{16}IB_{65}(100)$ containing 2E4MIm/TFSI (right panel). Reproduced with permission from ref 42. Copyright 2018 American Chemical Society.

morphologies in polymer electrolytes to improve ion transport properties.[28]

To expand these findings to other charged polymers and electrostatically control their thermodynamics, we varied the type of acid moiety. For this, SS_nMB_m, poly(styrenesulfonyl-(trifluoromethanesulfonyl)imide)-*b*-poly(methylbutylene) (ST_nMB_m), and poly(styrenephosphonate)-*b*-poly-(methylbutylene) (SP_nMB_m) block copolymers were synthesized having $-SO_3H$, $-NH(SO_2)_2CF_3$ (TFSI), and $-PO_3H_2$ groups, respectively.[14,23] To modulate the strength of electrostatic interactions in these polymers, a set of nonstoichiometric imidazolium ILs was introduced, where charged SS_n, ST_n, and SP_n chains have different binding affinities with imidazolium cations.

Figure 3a compares the phase diagrams of $SS_{11}MB_{17}(35)$ and $ST_{11}MB_{17}(35)$ containing nonstoichiometric 2E4MIm/HTFSI.[23] Neat $SS_{11}MB_{17}(35)$ and $ST_{11}MB_{17}(35)$ formed LAM and gyroid structures, respectively. Intriguingly, unprecedented reentrant phase transitions from LAM-to-A15-to-LAM-to-A15 were found for $SS_{11}MB_{17}(35)$ samples upon slight changes in the f values of ionic phases, unlike the sequential morphological changes from LAM-to-HEX-to-A15 of $ST_{11}MB_{17}(35)$ analogues. The A15 phases of the $SS_{11}MB_{17}(35)$ samples were stabilized for ILs containing excess [2E4MIm], whereas increase the content of [HTFSI] yielded weakly segregated LAM structures.

The A15 structure is a Frank–Kasper phase, and the stability window of this phase is narrow for block copolymers and unprecedented for charged block copolymers. Figure 3b shows the micellar morphology and radial distribution of each species in 2E4MIm/HTFSI, as obtained by molecular dynamics (MD) simulations. As shown, the PSS chains form a thin ionic shell layer upon interaction with $2E4MIm^+$ cations, but this was not observed for the $ST_{11}MB_{17}(35)$ analogues. In addition, as shown by density functional theory (DFT) calculations, the electrostatic interactions of $2E4MIm^+$ and $-SO_3^-$ are stronger than that with the $-TFSI^-$. Therefore, in the former case, enhanced electrostatic interactions at the micellar interface are key to stabilizing the distinctly shaped micelles in the A15 structure. In particular, the MB_m chains in the micellar core are stretched, whereas the SS_n chains in the shell are collapsed to yield conformational asymmetry of 1.84, consistent with the literature value of 1.85 reported for the A15 phases of neutral diblock copolymers. This value was slightly smaller (1.68) for $ST_{11}MB_{17}(35)$.[51]

Overall, the ion transport properties of the A15-forming samples with major ionic domains were superior to LAM counterparts. For example, for the $SS_{11}MB_{17}(35)$, a radical enhancement in the conductivity by an order of magnitude was seen if the samples formed A15 structures with the use of 2E4MIm-excessive ILs. Representative morphology factors obtained from the $SS_{11}MB_{17}(35)$ samples are presented in Figure 3c. For the A15 structures, morphology factors close to

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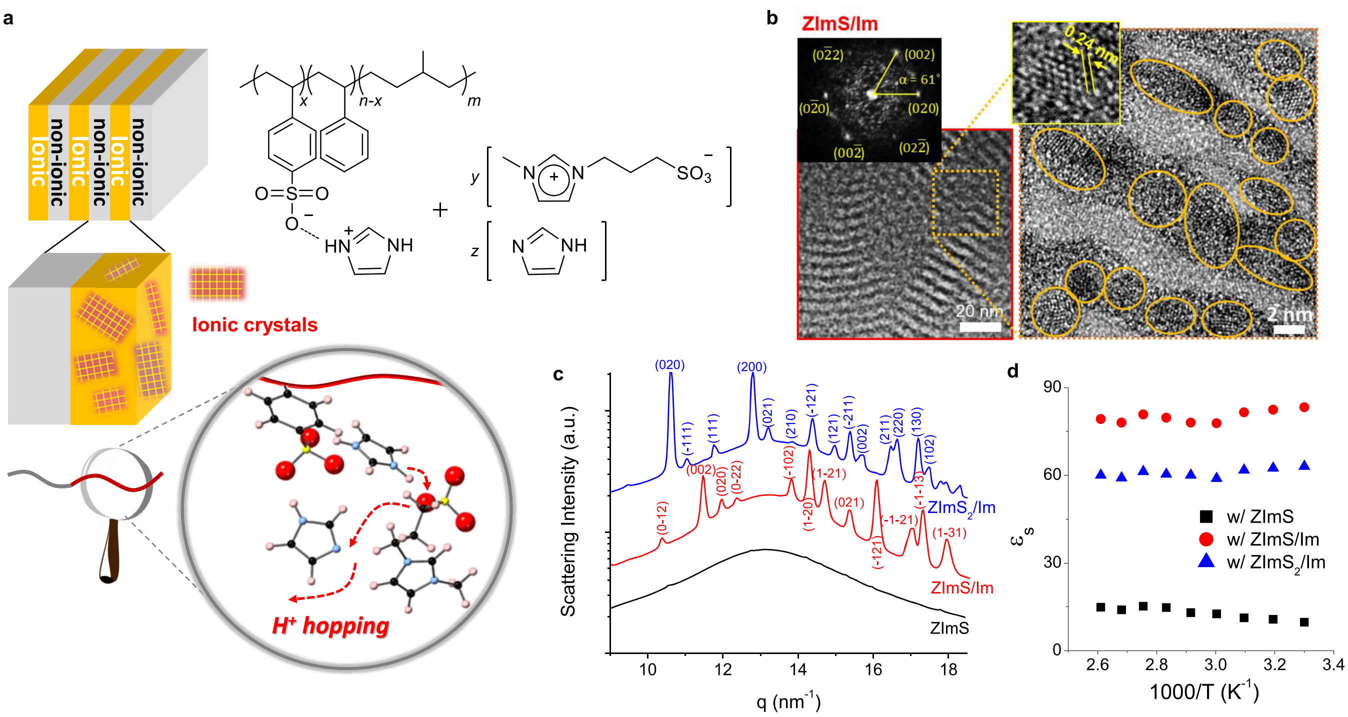

Figure 5. (a) Schemes depicting the facilitated proton hopping through hierarchically ordered ionic crystals in single-ion conducting block copolymers. (b) WAXS profiles and (c) static dielectric constant of $SS_{10}MB_{14}(20)$ comprising ZImS, ZImS/Im, and $ZImS_2$/Im. (d) Representative TEM images and FFT analysis of $SS_{10}MB_{14}(20)$ with ZImS/Im, revealing ionic crystals in SS domains. Reproduced with permission from ref 3.

1.0 were obtained, whereas that of the LAM structure was as low as 0.01. These results suggest routes to develop efficient block copolymer electrolytes having electrostatically controllable interfaces to exploit unexplored morphologies.

We now compare the phase behavior of IL-containing SS_nMB_m and SP_nMB_m.[14] The SAXS profiles in Figure 4a indicate a gyroid morphology for $SS_{17}MB_{26}(32)$, while $SP_{17}MB_{26}(40)$ was disordered. Therefore, $SP_{17}MB_{26}$ block copolymers have χ values lower than those of their $SS_{17}MB_{26}$ analogues, probably because the SP chains are less hydrophilic than those in the SS counterparts.[52] In addition, DFT calculations revealed that the $-PO_3H_2$ moieties remain protonated in the presence of imidazole (forming hydrogen bonds in the neutral state) because of its relatively low acidity, whereas the deprotonated $-SO_3^-$ anion can interact electrostatically with imidazolium cations.

Intriguingly, although the high T_g of the SP_n chains lowered the ionic conductivity of IL-containing SP_nMB_m to a greater extent than those of the SS_nMB_m counterparts, the hydrogen-bonded networks formed of the diprotic $-PO_3H_2$ moieties in SP_n domains increased the proton transference number of SP_nMB_m (Figure 4b). This implies that proton transport in SS_nMB_m and SP_nMB_m is fundamentally different.[14]

Various self-assembled morphologies were obtained for IL-containing SP_nMB_m by controlling the phosphonation level (PL), nonstoichiometry of the ILs, and IL content.[22,25] The phase diagram (Figure 4c) shows the nonstoichiometry of the ILs to be the key factor determining the morphology. For instance, the addition of excess $2E4MIm^+$ to $TFSI^-$ results in highly segregated A15 structures for both $SP_{10}MB_{14}(23)$ and $SP_{16}MB_{23}(38)$, whereas increasing the $TFSI^-$ content disrupts the ordered morphology. Intriguingly, A15 structures are common in acid-tethered block copolymers comprising nonstoichiometric ILs with excess cation contents. The exploration of the structure−transport relationship of IL-containing SP_nMB_m revealed higher morphology factors (0.83−0.96) for the A15 structures than for HEX structures (0.4).

All aforementioned acid-tethered polymers are randomly charged polymers. We thus carried out controlled polymerization of acid-tethered monomers to prepare new sets of polymers with 100% substitution degree. We also designed a monomer having two $-PO_3H_2$ groups at the 3- and 5-positions of the styrene repeating unit (poly(styrene bisphosphonate), SbP_n) and one at the 4-position (SP_n) as a control.[42] Block copolymers were synthesized using a polyisobutylene (PIB) macroinitiator, and these are denoted as poly(styrenephosphonate)-*b*-poly(isobutylene), $SP_nIB_m(100)$, for the monophosphate and poly(styrene bisphosphonate)-*b*-poly(isobutylene), $SbP_nIB_m(100)$, for the bisphosphonate.

$SbP_nIB_m(100)$ displayed well-defined nanostructures in both the absence and presence of ILs, unlike previously reported highly phosphonated polymers. As shown in Figure 4d, the SAXS profiles of IL-containing $SbP_{28}IB_{65}(100)$ (chemical structure is given in inset) at $\lambda = 0.5$, that is, one IL per SbP repeating unit, indicate LAM structures and segregation strengths increasing in order [Im][TFSI], [2MIm][TFSI], and [2E4MIm][TFSI] resulting from the hydrophobicity of the SbP_n chains and increased hydrophobicity of the alkyl-substituted imidazolium cations.

Intriguingly, although the $SbP_nIB_m(100)$ samples having various molecular weights and block compositions formed highly segregated ordered structures, the monosubstituted

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counterparts, $SP_nIB_m(100)$, had ill-defined morphologies (see the right-hand panel of Figure 4d). This difference originates from highly suppressed ion aggregation in the SbP_n domains despite the doubled local concentration of acid moieties, which enabled thermodynamically controlled microphase separation.[42] In contrast, the ion agglomeration in the SP_n domains disrupts the block interfaces.

This affects the ion transport properties, and the $SbP_nIB_m(100)$ samples showed 2–4-times higher ionic conductivities than those of the $SP_nIB_m(100)$ analogues owing to the low energy barrier for ion conduction in the virtually homogeneous SbP domains. Given that the T_g of the SbP chain is higher (>20 °C) than that of the SP chain, this approach improved ionic conductivity without sacrificing mechanical strength toward advanced polymer electrolytes.

Single-Ion Conducting Block Copolymers Having Hierarchically Ordered Ionic Crystals

For future electrochemical/electromechanical devices, single-ion polymers with limited anion depletion are promising but suffer from conductivities lower than those of conventional polymer electrolytes.[2,53,54] We used zwitterions rather than ILs in SS_nMB_m block copolymers as polar but nonionic additives, and neutral imidazole (Im) was also employed with various zwitterion:Im molar ratios to control the dipolar interactions with the SS chains.[3]

Remarkably, hierarchically ordered ionic crystals were formed within the ionic domains of the LAM structures of $SS_{10}MB_{14}(20)$ via the self-organization of the ZImS, Im, and SO_3^-–Im^+ pair (Figure 5a). Subsequently, ionic crystal formation could be regulated by varying the ZImS:Im molar ratio, which tuned the distance between neighboring proton donors and acceptors and the dipolar orientation. Bright-field TEM images, high-resolution TEM images, and fast Fourier transform (FFT) analysis in Figure 5b show the LAM morphology with 2–3 nm crystalline structures confined in SS domains (see circles) in $SS_{10}MB_{14}(20)$ containing ZImS/Im. WAXS profiles (Figure 5c) confirm the formation of crystals in the triclinic *P1* and monoclinic $P2_1/c$ space groups for $SS_{10}MB_{14}(20)$ containing ZImS/Im and $ZImS_2$/Im, respectively, but no crystal formation by adding ZImS alone. Promisingly, the static dielectric constant of $SS_{10}MB_{14}(20)$ was increased to 78 with ZImS/Im (Figure 5d) because of the well-arranged dipoles in the SS nanodomains with densely packed ionic crystals, which was not shown with ZImS addition.[3] This indicates that the ion pair (SO_3^-–Im^+) in $SS_{10}MB_{14}(20)$ and added ZImS tend to form a quadrupole, which can be broken with the coaddition of neutral Im to allow the close packing of dipolar moieties, leading to the unprecedented ionic crystal formation.

Furthermore, the proton diffusion coefficient obtained for $SS_{10}MB_{14}(20)$ with ordered ionic crystals was 2–5-times higher than those of its amorphous counterparts. Irrespective of the ratio of ZImS/Im, the SS_n homopolymers produced amorphous WAXS profiles and no improvement in ionic conductivity nor increase in dielectric constant, indicating that nanoconfinement by SS_nMB_m stabilized the entropically unfavorable ionic crystal formation. This work represents the first report of high-conductivity, high-dielectric constant, and leak-free single-ion conducting polymers.

Electroactive Polymer Actuators Based on Charged Block Copolymers

As next-generation technologies based on charged polymers, soft EAP actuators have been investigated to redefine human–machine interactions using artificial muscles, although they are still in the nascent stage.[55] Although dielectric EAPs require kilovolt stimulus,[56] *i*EAPs respond to potentials of 1–3 V,[2,30] enabling high signal throughput with portable power supplies. *i*EAP actuators function because ion movement induces asymmetric volume changes in the polymer layers, and thus, their ion transport properties are the key to determining actuation performance. Nafion, poly(vinylidene fluoride-*co*-hexafluoropropylene) (PVdF(HFP)), and a few block copolymers have been used in *i*EAP actuators, but achieving low-voltage operation, rapid responses, and high bending without back-relaxation is challenging.[57,58]

In 2013, we reported unprecedented *i*EAP actuation performance based on charged block copolymers.[30] The actuator was fabricated with a trilayer structure in which a 50–100 μm-thick polymer layer is sandwiched between ~20 μm-thick carbon nanotube electrodes. By embedding quaternary ILs in SS_nMB_m with tunable ionic interactions and ion sizes, we prepared a low-voltage *i*EAP with better performance than previously reported polymer actuators (Figure 6a). Unique HEX-structured block copolymers having ion concentration gradients beneath the cathode during actuation yielded a bending strain of 4% without back-relaxation and rapid millimeter-scale displacements (<1 s) under sub-1-V con-

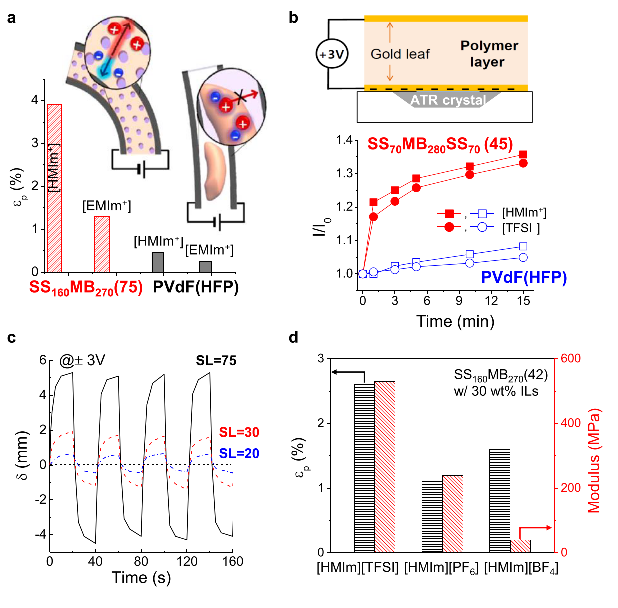

Figure 6. (a) Improved bending strain of $SS_{160}MB_{270}(75)$/IL actuators, compared with PVdF(HFP) counterparts owing to well-connected ionic domains, as schematically illustrated in insets. The type of IL cation largely affects the bending strain. (b) In situ monitoring of ion migration across $SS_{70}MB_{280}SS_{70}(45)$ and PVdF-(HFP) membranes toward the negative electrode surface. (c) SL-dependent bending displacement of SS_nMB_m/IL actuators. (d) Bending strain and modulus of $SS_{160}MB_{270}(42)$ with different ILs. (a, c) Reproduced with permission from refs 30 and 31. Copyright 2013 Springer Nature and Copyright 2014 American Chemical Society. (b, d) Reproduced with permission from ref 31. Copyright 2014 American Chemical Society.

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ditions. In contrast, the discrete ionic domains of PVdF(HFP) membranes deteriorate ion diffusion (see inset). To enhance the generated stress, we also used $SS_nMB_mSS_n$ triblock copolymers, yielding 2−3 times greater moduli than those of the diblock analogues.

As shown in Figure 6b, in situ monitoring of ions near the electrode surfaces by attenuated total reflection spectroscopy revealed the coupled motion of IL cations and anions in $SS_nMB_mSS_n$ membranes at given voltages, where $HMIm^+$ is the charge carrier.[31] Time-resolved intensity ratios of $HMIm^+$ and $TFSI^-$ at the cathode surface increased significantly on the application of 3 V for $SS_nMB_mSS_n$/ILs compared to the modest increase observed for PVdF(HFP) analogues. In situ SAXS experiments on $SS_nMB_mSS_n$/ILs confirmed noticeably increased domain sizes near the negative electrode surface arising from the migration of both $HMIm^+$ and $TFSI^-$.[31]

Further, the actuator bending strain improved upon increasing the SL of SS_nMB_m (Figure 6c) because this enhanced ion dissociation and lowered the activation energy for ion conduction. Another key factor regulating the bending strain of *i*EAP actuators is the stress generated in the polymer layer at a given deformation, which is significantly affected by the IL. As shown in Figure 6d, despite the lowest viscosity of [HMIm][TFSI], it yielded the highest modulus when used in SS_nMB_m owing to the effective intermolecular interactions with SS chains.[31] By taking advantage of inherently high conductivity of [HMIm][TFSI], the actuation performance was enhanced.

We next developed soft actuators with rapid, reversible responses and ultrafast millisecond-scale mechanical deformations at 1 V in air without fatigue.[2] For this, the IL-containing polymer layer was replaced by single-ion conducting polymers prepared from Im-doped $SS_{153}MB_{313}(60)$, which enabled cation diffusion across SS domains while retaining the mechanical integrity of the ionophobic MB cylinders. To overcome the low ionic conductivity of single-ion polymers, the aforementioned zwitterions having sulfonated anions were used by varying the cations (Figure 1d) with fine-tuned degree of intermolecular interactions in the SS domains. Figure 7a shows a schematic of the structure and ion transport of single-ion actuators based on the Im-doped $SS_{153}MB_{313}(60)$ containing zwitterions.[2]

Remarkably, our single-ion actuators with ZImS yielded 1.8% bending strain at 3 V. Bending in IL-based actuators takes 5 s, but the ZImS-based actuator responded in 20 ms, moving 1 mm in 60 ms and achieving a fully bent position after 190 ms at 1 V (Figure 7b), which is unprecedented, with three-times greater bending strain.[2] The single-ion actuators also showed improved blocking force and in-air stability over 20 000 cycles without back-relaxation, which makes them promising for applications in cutting-edge biomimetic technologies. The impressive performance can be ascribed to the improved ionic conductivity of ZImS-doped $S_{153}MB_{313}(60)$, which surpasses those of IL-containing membranes (both cations and anions are conducting) (Figure 7c).

Figure 7d shows the increased static dielectric constants of $SS_{153}MB_{313}(60)$ from 7 to 76 upon adding ZImS, suggesting that the Im^+ cation easily dissociates under synergistic dipole alignment to increase the charge density. As a result, the bending strain increased and the switching speed was noticeably faster, compared to those of PVdF(HFP) and Nafion-based actuators (inset).

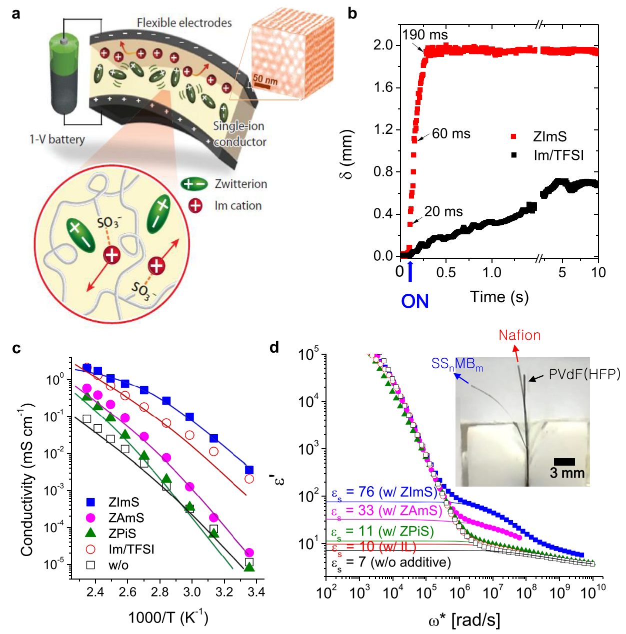

Figure 7. (a) Schematic illustration of the structure and ion transport of single-ion *i*EAP actuator fabricated with zwitterion-containing $SS_{153}MB_{313}(60)$. (b) Time-dependent displacement of ZImS-containing single-ion polymer actuators, compared with IL-containing analogue under 1 V. (c) Ionic conductivity and (d) dielectric constant of Im-doped $SS_{153}MB_{313}(60)$ with added zwitterions. Reproduced with permission from ref 2. Copyright 2016 The Authors. Published by Springer Nature under a Creative Commons Attribution License (CC BY 4.0) http://creativecommons.org/licenses/by/4.0/.

Low-Power Soft Actuators as Artificial Muscles and Wearable Electronics

To combine artificial structures with living systems and mimic or assist biological processes, we have explored various biohybrid actuators. For example, we used our superfast single-ion actuator as fully functional artificial muscles.[32] By interfacing the actuator with an organic synapse (Figure 8a), the charged block copolymers could be used to measure voltage signals from light-sensitive artificial synapses. These synaptic behaviors are similar to the activation responses of skeletal muscles that contract proportionally to the frequency of action potentials received from motor neurons. Because these voltage signals are extremely low, *i*EAP actuators can readily mimic the contractile response triggered by presynaptic action potentials received from light-sensing devices in postsynaptic signal form. The voltage of around 1 V generated from the artificial synapse before illumination resulted in the minimal contraction of the *i*EAP actuator, which then increased with the number of voltage spikes upon applying short pulses of light, as shown in Figure 8b, indicating their potential for use as artificial muscles in a biological setting.

Like plants and other living organisms, adaptable materials can respond to stimuli such as voltage, light, humidity, temperature, pH, and solvent.[59] Thus, we developed dual-responsive actuators by combining a light-active polymer (LAP) layer with an *i*EAP actuator, denoted the LEAP actuator, as depicted in Figure 8c.[33] LAP is based on ionically cross-linked random copolymers with azobenzene moieties and undergoes rapid *trans*-to-*cis* isomerization under UV irradiation

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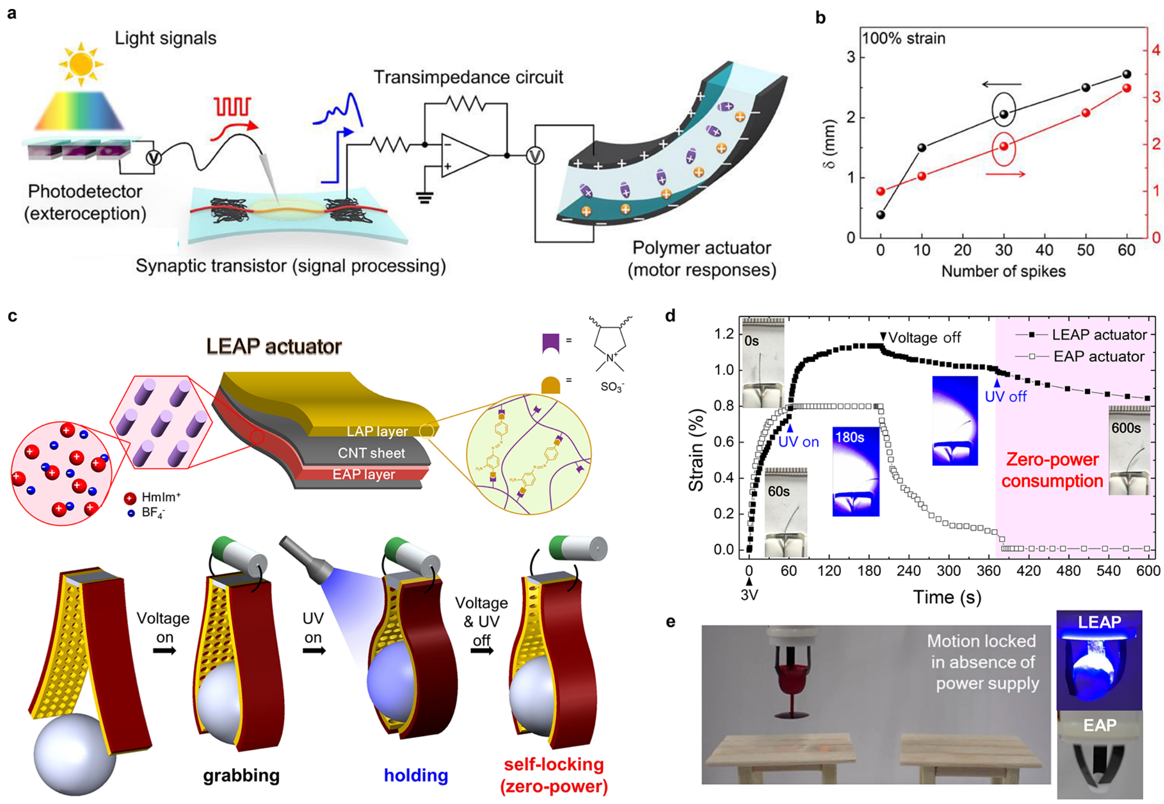

Figure 8. (a) Configuration of organic optoelectronic synapse and neuromuscular electronic system. (b) Maximum displacement of actuator and output voltage with number of spikes. Reproduced with permission from ref 32. Copyright 2018 AAAS. (c) LEAP actuator comprising light-active and electroactive polymer layers enabled grabbing, holding, and self-locking behaviors. (d) Time-dependent bending strain of LEAP and EAP actuators at 3 V under UV irradiation. (e) LEAP actuators holding a wine glass without power supply and that can hold heavier objects than EAP actuators. Reproduced with permission from ref 33. Copyright 2018 Wiley-VHC.

with slow *cis*-to-*trans* reversal after the light source is removed. The thicknesses of the LAP layer, EAP layer, and carbon nanotube electrode were ∼15, ∼30, and ∼10 μm, respectively. The LEAP actuator is capable of grabbing (voltage response), holding (UV response), and self-locking (in the absence of power supply) (Figure 8c), offering controlled deformations, programmability, and site-specific actuation.

To improve its grip, various micropatterns were formed in the LAP layer, and the hole pattern yielded the highest bending strain via the effective localization of the load at the pattern boundaries. Further, self-locking without stimulus was used to hold a wine glass for 3 min (Figure 8e, input power of only 6.2 mW h cm^{-2}). In addition, the LEAP actuator held an object three times heavier than what was possible using the *i*EAP actuator (Figure 8e, right-hand inset)

SUMMARY AND OUTLOOK

In this Account, we have presented our work on charged block copolymers, focusing on controlled thermodynamics, nanoscale self-assembly, and long-range interactions. Key factors in this research area are the type of ionic additive and Coulombic interactions between ions and polymer functional groups. By understanding these interrelated factors, conductive and robust polymer electrolytes can be prepared. The development of acid-tethered polymers is one route to prepare elaborate nanostructures and establish new structure−transport relationships. Furthermore, by varying the ionic moieties, changing the local concentration of ions, and developing ionic-crystal-based single-ion polymers, the ion transport properties and morphologies of charged block copolymers can be fine-tuned. We conjecture that the future avenue of charge block copolymers is bifunctional polymer-based electrolytes in which two different functional moieties are bound to precise positions in the backbones. The main chain structures and the functional groups should be designed based on theoretical predictions for a concrete understanding of ion distribution at distances of several angstroms, ion aggregation at several nanometers, and microphase separation at a few tens of nanometers to elucidate ion transport across multiscale self-assembled structures.

In addition, fundamental studies on charged polymers could also produce advanced devices such as energy-efficient electromechanical wearable technologies and artificial muscles. Soft actuators based on charged block copolymers can be powered by lightweight power accessories exploiting programmable deformations with significant force generation applicable to next-generation technology. The further introduction of different combinations of multistimuli-responsive polymers could enable the development of complex systems mimicking artificial muscles and haptic devices as well as the complex deformations observed in nature.

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